# Professor Assistant
# Program

Automated Exam Generation from a Question Bank

*Team MANIAC | Python Final Project*

# Project Overview

| | |
|---|---|
| **Goal** | Help professors automatically generate exams from a pre-built question bank |
| **Language** | Python — uses only the built-in random module |
| **Input** | question_bank.txt with Q&A pairs |
| **Output** | A saved exam file with randomly selected questions |

# Team MANIAC

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# Program Workflow

1 **Start**

Professor launches the program

2 **Enter Name**

Validates letters-only input

3 **Load Bank**

Reads question_bank.txt Q&A pairs

4 **Set Count**

Professor picks number of questions

5 **Generate**

Randomly selects unique questions

6 **Save Exam**

Writes exam to chosen output file

# Key Functions

## `read_question_bank()`

Reads question_bank.txt, strips whitespace, validates even-line format, and returns a list of (question, answer) tuples. Returns None on errors.

## `create_exam()`

Randomly selects unique questions using index tracking. Writes numbered Q&A pairs to the output file. Caps selection at available count.

## `main()`

Controls program flow: validates professor name, loops exam creation, and displays a personalized farewell on exit.

# Question Bank Format

- Plain .txt file — one question per line, answer on the next line
- Lines come in pairs: Question → Answer → Question → Answer
- Blank lines are automatically skipped during processing
- Odd number of lines signals an invalid file — returns None

```
sample  question_bank.txt



Who is the father of Computer science?
Charles Babbage

What does CPU stand for?
Central Processing Unit

BIOS stands for?
Basic Input Output System
```

*Contains 100 Q&A pairs on computer science fundamentals*

# Input Validation

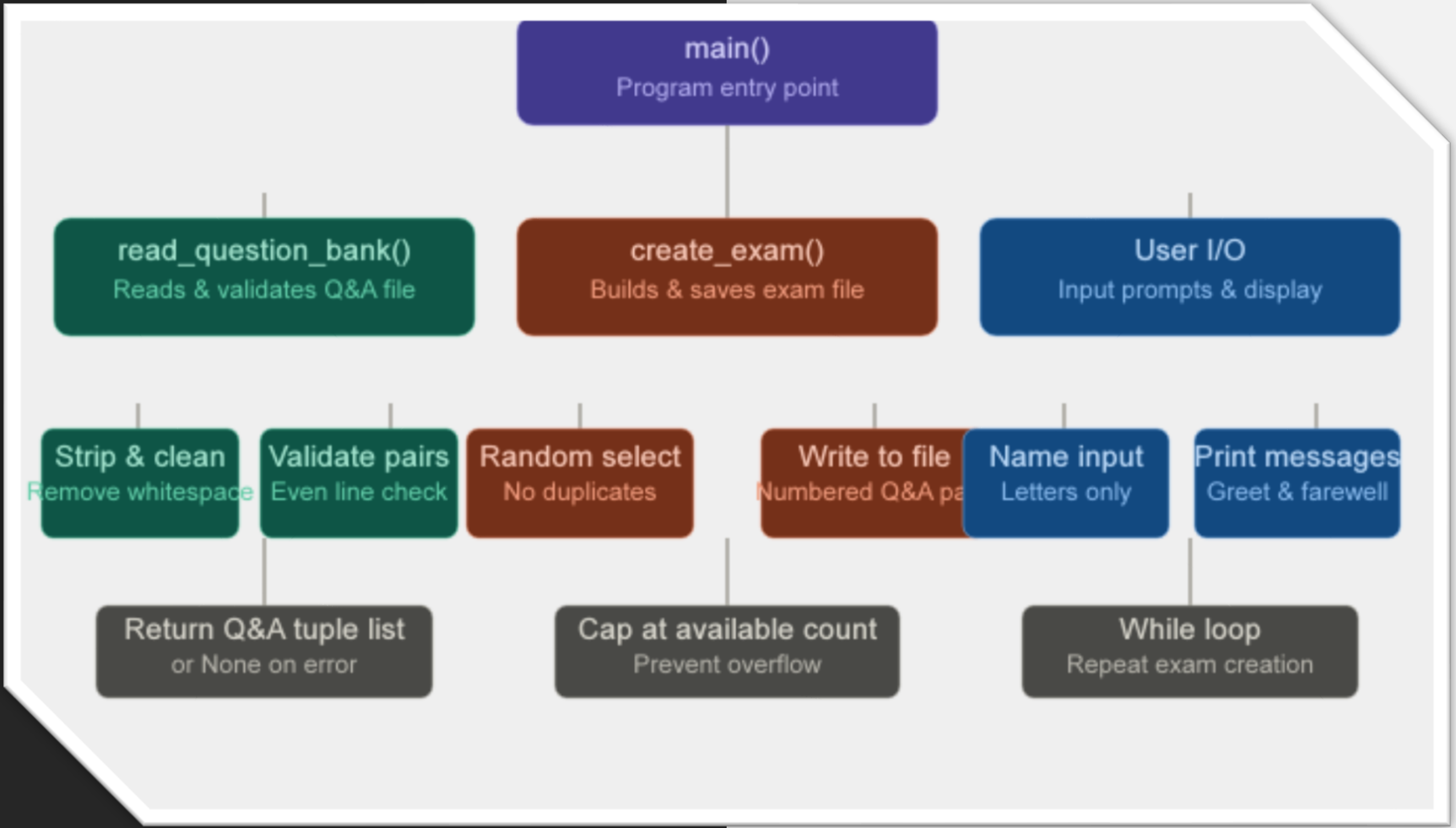
| | |
|---|---|
| **Professor Name** | Must contain letters only. Spaces are stripped before check. Loop repeats until valid input is given. |
| **Question Count** | If count exceeds available questions, it is automatically capped at the bank size — no crash. |
| **File Existence** | read_question_bank() catches FileNotFoundError and returns None, prompting a user-friendly error message. |
| **File Format** | Odd number of cleaned lines returns None — prevents incorrect Q&A pairing. |

```
Welcome to professor assistant version 1.0.
Please Enter Your Name: Ibrahim Musa
Hello Professor Ibrahim Musa , I am here to help you create exams from a question_bank.txt file.
Do you want me to help you create an exam (Yes to proceed | No to quit the program)? yes
Yes, indeed the path you provided includes questions and answers.
How many question-answer pairs do you want to include in your exam? 30
Where do you want to save your exam? exam
Congratulations Professor Ibrahim Musa .
Your exam is created and saved in exam.
```

# Sample Program Output

# Challenges & Solutions

| CHALLENGES | SOLUTIONS |
| --- | --- |
| Preventing duplicate questions | Track used indexes in a list; loop until new index found |
| Handling missing or malformed files | try/except FileNotFoundError + even-line format check |
| Capping question count safely | Cap total_questions to available_questions automatically |
| Keeping name input valid | Use .replace(' ','').isalpha() with a retry loop |


main()
Program entry point
read_question_bank()
Reads & validates Q&A file
create_exam()
Builds & saves exam file
User I/O
Input prompts & display
Strip & clean
Remove whitespace
Validate pairs
Even line check
Random select
No duplicates
Write to file
Numbered Q&A pa
Name input
Letters only
Print messages
Greet & farewell
Return Q&A tuple list
or None on error
Cap at available count
Prevent overflow
While loop
Repeat exam creation

# Conclusion

- Automates exam creation — saves professors significant time
- Robust error handling for missing files and bad input
- Guarantees no duplicate questions per exam session
- Simple, clean Python design using only the random module
- Scalable: question bank can be expanded at any time